

SMART INDIA HACKATHON 2026

SIH26045

IP-SAKTI Sahayak

A multilingual, RAG-based, source-cited AI assistant for Intellectual Property and regulatory guidance in Ayurveda

Ministry of Ayush | MedTech / BioTech / HealthTech | Software

Team Presentation Document

Prototype + Solution + Technical Architecture + Implementation Plan

1. Problem Statement

Ayurveda innovators, practitioners, researchers, AYUSH startups/MSMEs and cultivators often need to navigate multiple intellectual-property, traditional-knowledge, biodiversity and product-regulatory regimes at the same time.

- Choosing the correct protection route: patent, GI, trademark, copyright, design, trade secret or other applicable mechanism.
- Understanding whether a formulation is classical, proprietary/non-classical, a new drug, phytopharmaceutical, Ayurveda-Aahar/nutraceutical or cosmetic.
- Handling traditional knowledge, biological-resource access and benefit-sharing (ABS) considerations.
- Understanding India-specific requirements separately from international/export-market requirements.
- Finding authoritative, current and traceable sources instead of relying on generic AI answers.

Core challenge: The user needs a guided pathway from an Ayurveda idea/formulation to evidence-backed IP and regulatory next steps.

2. Proposed Solution

IP-SAKTI Sahayak is designed as a multilingual AI decision-support assistant that combines retrieval-augmented generation (RAG), a curated/version-tracked knowledge base and source citations.

- Ask questions in natural language and receive answers grounded in selected authoritative documents.
- Switch between India and international jurisdictions to avoid mixing legal/regulatory contexts.
- Classify an Ayurveda product before suggesting an IP/regulatory pathway.
- Generate an IP pathway checklist with documents, questions and next actions.
- Provide ABS/traditional-knowledge checks and pointers to relevant prior-art/TK resources.
- Show citations, confidence and an escalation route to a human IP/regulatory facilitator.
- Support multilingual text and voice interaction for wider accessibility.

3. End-to-End User Workflow

Step	User action	System response
1	Login / choose role and language	Personalized dashboard
2	Enter product/formulation details	Initial classification inputs
3	Choose India or International	Jurisdiction-specific retrieval
4	Ask IP / regulatory question	RAG retrieves relevant sources
5	Classify product	Relevant category + regulatory/IP considerations
6	Run IP Pathway	Checklist of protection/compliance steps
7	Run ABS / TK check	Flags information that may require further review
8	Review Sources	Citation-backed evidence
9	Escalate / export	Human review or structured report

4. Key Features

Dashboard — Product status, jurisdiction, workflow progress, pending reviews and quick actions.

Ask IP-SAKTI — Conversational interface for IP and regulatory questions with source-backed responses.

Product Classifier — Structured questions to determine the relevant Ayurveda product category.

IP Pathway — Guided checklist across applicable IP protection routes and next actions.

ABS / TK Check — Prompts for biological-resource and traditional-knowledge considerations.

Sources — Searchable source cards with document title, jurisdiction and citation context.

Voice Assistant — Browser-based voice interaction for accessible question input.

Multilingual UI — Language selection designed for broader AYUSH ecosystem access.

Human Review — Escalation when the system lacks sufficient evidence or the issue needs professional review.

5. Technical Architecture

The proposed full system is designed as a source-grounded RAG pipeline rather than a free-form chatbot.

Layer	Components
User Interface	Responsive web app, dashboard, chat, forms, multilingual and voice UI
Application API	Authentication, user/session management, orchestration and workflow services
RAG Layer	Document ingestion, chunking, embeddings, retrieval, reranking and grounded generation
Knowledge Base	Version-tracked statutes, rules, treaties, standards, registry/official records and curated guidance
Knowledge Graph	Relationships among IP types, product categories, jurisdictions, authorities, documents and requirements
Safety Layer	Citation checks, confidence, safe abstention, privacy/audit controls and human escalation
Data Layer	Relational database + vector store; graph database where needed

Suggested stack: React/Next.js + Tailwind CSS; FastAPI; LLM; embeddings + reranker; PostgreSQL; pgvector/Qdrant/Weaviate; Neo4j where graph relationships add value; OCR/PDF ingestion; secure authentication; Docker/cloud deployment.

6. RAG & Source-Citation Design

- Retrieve only from the selected jurisdiction and relevant topic before generating an answer.
- Attach document title, authority, version/date and relevant citation context to each important answer.
- Use confidence thresholds and abstain when evidence is insufficient or conflicting.
- Keep the legal/regulatory corpus version-tracked so updates can be audited.
- Separate information from legal advice and show a standing disclaimer.
- Route complex/high-risk questions to a human IP/legal/regulatory facilitator.

Important prototype boundary: the demonstration UI can use curated/mock source records; a production deployment must connect the RAG layer to an authenticated, maintained corpus of authoritative sources.

7. Innovation / Differentiation

- Source-cited conversational assistance instead of an uncited general chatbot.
- Formulation classification is placed before pathway recommendations.
- India vs International mode reduces jurisdiction mixing.
- IP + regulatory + ABS/TK considerations are brought into one workflow.
- Version-tracked evidence and safe abstention support reliability.
- Multilingual + voice interaction improves accessibility.
- Human escalation is built into the workflow rather than treating AI output as final legal advice.

8. Prototype Demonstration

The prototype is intended to be demonstrated as a clickable journey rather than only as static slides.

Demo sequence:

- Login → select role and language.
- Open Dashboard → view product/jurisdiction status.
- Ask IP-SAKTI → ask an Ayurveda IP/regulatory question.
- Product Classifier → enter formulation/product details.
- IP Pathway → inspect suggested checklist.
- ABS Check → review biological-resource/TK prompts.
- Sources → inspect evidence cards.
- Voice → demonstrate voice input where browser permissions allow.

The prototype should be presented as a functional concept demonstration; it should not be described as a substitute for professional legal advice or as a live authoritative legal database unless those integrations are actually deployed.

9. Feasibility & Implementation Plan

Phase	Deliverable
Phase 1 — MVP	Citation-grounded RAG, core Indian IP/regulatory corpus, dashboard, classifier and source display.
Phase 2	Knowledge graph, ABS/TK workflow, better routing, confidence and human escalation.
Phase 3	Multilingual/voice expansion, international jurisdiction packs and connectors.
Phase 4	Continuous corpus updates, evaluation, security hardening and production deployment.

Evaluation: answer accuracy, citation correctness, retrieval quality, safe-abstention rate, multilingual quality, latency, usability and human-review outcomes.

10. Expected Impact

- Reduce the effort required to locate and interpret relevant IP/regulatory information.
- Improve traceability by keeping source evidence attached to AI responses.
- Support AYUSH innovators and institutions in earlier-stage IP and compliance planning.
- Increase awareness of traditional-knowledge and biodiversity/ABS considerations.
- Provide a common interface across multiple information regimes and languages.
- Create a foundation for a maintainable national-level knowledge service, subject to appropriate authority, governance and expert oversight.

11. Reference & Governance Notes

The original uploaded SIH material for the user's crop-health project is a six-page SIH presentation with the structure: title/problem statement, proposed solution, technical approach, feasibility/viability, impacts/benefits and research/references.

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This new document follows a similar presentation-document structure but is prepared for SIH26045 - IP-SAKTI Sahayak, the Ministry of Ayush problem statement supplied in the conversation.

Governance principle: AI output is decision support, not legal advice. Production use requires authoritative source maintenance, versioning, security, auditability, professional review pathways and appropriate legal/regulatory governance.

END OF PRESENTATION DOCUMENT